

Interval Belief Propagation for Credal Networks with Multi-Valued Variables

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Abstract

Interval Belief Propagation (IBP) is an iterative message-passing algorithm for approximate marginal inference in credal networks. Unlike the 2U and L2U algorithms which are restricted to binary variables, IBP operates natively on multi-valued variables by propagating interval-vector messages. At each iteration, variable nodes tighten their interval bounds by intersecting incoming messages, while factor nodes compute new bounds by optimizing over the extreme points of local credal sets and the corner points of incoming interval messages. The algorithm is inspired by the ApproxLP framework of Antonucci et al. (2015) and the modified belief propagation for Logical Credal Networks (Qian et al., 2021).

1 Preliminaries

1.1 Factor Graph

Given a credal network $\langle G, \mathcal{K} \rangle$, we construct a **factor graph**: a bipartite graph with variable nodes and factor nodes. Each variable X_i in the credal network corresponds to a variable node with cardinality $|D_i|$. Each conditional credal set $K(X_i \mid \text{Pa}(X_i))$ defines a factor node f_i whose scope is $\{X_i\} \cup \text{Pa}(X_i)$. The factor stores the extreme points of $K(X_i \mid \pi)$ for all parent configurations π .

1.2 Interval Messages

Definition 1 (Interval Message). *For a variable X with k states, an **interval message** is a pair of vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$ where $0 \leq l_x \leq u_x \leq 1$ for each state $x \in D_X$. The interval $[l_x, u_x]$ bounds the probability $P(X = x)$ from the message sender's perspective.*

Messages are of two types:

- $\mu_{v \rightarrow f} = (\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$: message from variable v to factor f .
- $\mu_{f \rightarrow v} = (\mathbf{l}_{f \rightarrow v}, \mathbf{u}_{f \rightarrow v})$: message from factor f to variable v .

All messages are initialized to the vacuous interval $([0, \dots, 0], [1, \dots, 1])$.

2 Algorithm

Algorithm 1 Interval Belief Propagation (IBP)

Require: Factor graph with extreme points, query Z , evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , threshold τ

Ensure: Bounds $[P(Z=z), \overline{P}(Z=z)]$ for each z

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1: Initialize  $\mu_{v \rightarrow f} \leftarrow (\mathbf{0}, \mathbf{1})$  and  $\mu_{f \rightarrow v} \leftarrow (\mathbf{0}, \mathbf{1})$  for all edges
2: for each evidence variable  $Y_j = y_j$  do
3:   Set  $\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j})$  for all factors  $f \in \mathcal{N}(Y_j)$ 
4: end for

5: for  $t = 1, \dots, T$  do
6:    $\delta \leftarrow 0$ 
7:   for each variable  $v$  (not evidence), each factor  $f \in \mathcal{N}(v)$  do
8:     for each state  $x$  do
9:        $l_{v \rightarrow f}(x) \leftarrow \max_{f' \in \mathcal{N}(v) \setminus \{f\}} l_{f' \rightarrow v}(x)$ 
10:       $u_{v \rightarrow f}(x) \leftarrow \min_{f' \in \mathcal{N}(v) \setminus \{f\}} u_{f' \rightarrow v}(x)$ 
11:    end for
12:     $\mathbf{u}_{v \rightarrow f} \leftarrow \max(\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$  ▷ Ensure  $l \leq u$ 
13:    Update  $\delta$ 
14:  end for
15:  for each factor  $f_i$  (node  $X_i$ , parents  $\text{Pa}(X_i)$ ), each  $v \in \text{scope}(f_i)$  do
16:     $\mathbf{l}_{\text{new}} \leftarrow \mathbf{1}, \quad \mathbf{u}_{\text{new}} \leftarrow \mathbf{0}$ 
17:    for each vertex combination  $\mathbf{vc}$  of ext  $K(X_i \mid \cdot)$  do
18:      for each corner-point combination  $\mathbf{cc}$  of incoming intervals from  $\text{scope}(f_i) \setminus \{v\}$  do
19:        Compute unnormalized marginal  $\mathbf{m}(v)$  using  $\mathbf{vc}$  and  $\mathbf{cc}$  ▷ See Section ??
20:         $\mathbf{p} \leftarrow \mathbf{m} / \sum_x m(x)$ 
21:         $\mathbf{l}_{\text{new}} \leftarrow \min(\mathbf{l}_{\text{new}}, \mathbf{p})$ 
22:         $\mathbf{u}_{\text{new}} \leftarrow \max(\mathbf{u}_{\text{new}}, \mathbf{p})$ 
23:      end for
24:    end for
25:     $\mu_{f \rightarrow v} \leftarrow (\mathbf{l}_{\text{new}}, \mathbf{u}_{\text{new}})$ 
26:    Update  $\delta$ 
27:  end for
28:  if  $\delta < \tau$  then break
29:  end if
30: end for
31: for each state  $z$  do
32:    $\underline{P}(z) \leftarrow \max_{f \in \mathcal{N}(Z)} l_{f \rightarrow Z}(z)$ 
33:    $\overline{P}(z) \leftarrow \min_{f \in \mathcal{N}(Z)} u_{f \rightarrow Z}(z)$ 
34: end for
35: return  $[\underline{P}(z), \overline{P}(z)]$  for each  $z$ 

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2.1 Factor-to-Variable Message Computation

The factor-to-variable message from factor f_i (associated with node X_i , parents $\text{Pa}(X_i)$) to target variable $v \in \text{scope}(f_i)$ is computed as follows.

Let $\mathcal{O} = \text{scope}(f_i) \setminus \{v\}$ be the other variables. For each “other” variable $w \in \mathcal{O}$, let $(\mathbf{l}_w, \mathbf{u}_w) = \mu_{w \rightarrow f_i}$ be the incoming message.

Vertex enumeration. The factor stores extreme point vertices $V(\pi) = \text{ext } K(X_i \mid \pi)$ for each parent configuration π . A vertex combination selects one vertex per parent configuration.

Corner-point enumeration. For each variable $w \in \mathcal{O}$, the incoming interval $[l_w(x), u_w(x)]$ is evaluated at its two extreme values: $l_w(x)$ or $u_w(x)$. With $|\mathcal{O}|$ variables, there are $2^{|\mathcal{O}|}$ corner-point combinations.

Marginal computation. For a given vertex combination $\mathbf{vc}$ and corner-point assignment $\mathbf{cc}$, the unnormalized marginal over v is:

Case 1: $v = X_i$ (target is the child node).

$$m(v=x) = \sum_{\pi} P_{\mathbf{vc}}(X_i=x \mid \pi) \cdot \prod_{w \in \text{Pa}(X_i) \cap \mathcal{O}} q_w(\pi_w),$$

where q_w is the lower or upper bound vector selected by $\mathbf{cc}$ for variable w , and π_w is the value of w in configuration π .

Case 2: $v \in \text{Pa}(X_i)$ (target is a parent).

$$m(v=x) = \sum_{\pi: \pi_v=x} \left(\sum_{x_i} P_{\mathbf{vc}}(X_i=x_i \mid \pi) \cdot q_{X_i}(x_i) \right) \cdot \prod_{w \in \text{Pa} \setminus \{v\}} q_w(\pi_w),$$

where q_{X_i} is the incoming child message bound if $X_i \in \mathcal{O}$.

The marginal is then normalized: $p(x) = m(x) / \sum_{x'} m(x')$.

3 Convergence Properties

- Messages are updated by taking min/max over all vertex and corner-point combinations, so the algorithm explores the full range of the credal set.
- On polytree-shaped networks with binary variables, IBP reduces to the 2U algorithm and converges to exact bounds.
- On loopy graphs, convergence to a fixed point is empirically observed but not formally guaranteed. The fixed point may not correspond to the exact bounds (similar to loopy belief propagation for Bayesian networks).

4 Running Example

Example 1 (Alarm Network). *Using the same alarm network with variables B (2 states), E (2 states), A (2 states), and CD (4 states).*

Factor graph. Four factors: f_A (scope $\{A, B, E\}$), f_B (scope $\{B\}$), f_E (scope $\{E\}$), f_{CD} (scope $\{CD, A\}$).

Query: $P(B=1)$, **no evidence.** Variable B connects to factors f_B and f_A .

Iteration 0: All messages are vacuous $([0, 0], [1, 1])$.

$f_B \rightarrow B$: Factor f_B has scope $\{B\}$ with vertices $[0.8, 0.2]$ and $[0.9, 0.1]$. No other variables in scope, so: $l_{f_B \rightarrow B} = [0.8, 0.1]$, $u_{f_B \rightarrow B} = [0.9, 0.2]$. These are the correct marginal bounds for B from its local credal set.

$f_A \rightarrow B$: Factor f_A has scope $\{A, B, E\}$. With vacuous messages from A and E , the message to B is computed by enumerating vertex and corner-point combinations. Since A and E messages are vacuous $[0, 1]$, the resulting bounds for B are wide.

Final marginal for B : $\underline{P}(B=1) = \max(l_{f_B \rightarrow B}(1), l_{f_A \rightarrow B}(1))$, $\overline{P}(B=1) = \min(u_{f_B \rightarrow B}(1), u_{f_A \rightarrow B}(1))$. The bounds from f_B provide the tightest constraints: $[0.1, 0.2]$. The bounds from f_A are wider and do not further constrain the result.

Comparison with exact VE.

<i>Method</i>	$\underline{P}(B=1)$	$\overline{P}(B=1)$
<i>Exact VE</i>	<i>0.1000</i>	<i>0.2000</i>
<i>Interval BP</i>	<i>0.1000</i>	<i>0.2000</i>

For root nodes, IBP recovers the exact bounds because the local factor alone determines the marginal. For non-root nodes with evidence, the bounds may differ due to the approximation inherent in loopy message passing.

5 Multi-Valued Variable Support

A key advantage of IBP over L2U and GL2U is native support for multi-valued variables:

- **L2U:** Restricted to binary variables. Messages are scalar intervals.
- **GL2U:** Handles multi-valued variables by binarization (replacing a k -valued variable with $\lceil \log_2 k \rceil$ binary variables in a chain). This introduces auxiliary structure and can be lossy.
- **IBP:** Messages are interval vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$. No binarization needed. Each state is bounded independently.

For example, the compound variable CD in the alarm network has 4 states. IBP directly propagates 4-dimensional interval vectors for CD , while GL2U would require binarizing CD into two binary variables.

6 Complexity

Each iteration of IBP takes $O(N_f \cdot V_{\max} \cdot 2^{S_{\max}} \cdot k^{S_{\max}})$ time, where N_f is the number of factors, $V_{\max}$ is the maximum number of vertex combinations per factor, $S_{\max}$ is the maximum scope size, and k is the maximum domain size. The number of iterations until convergence depends on the network structure and is typically 10–100 for practical networks.

7 References

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